



**BAULKHAM HILLS HIGH SCHOOL**

**2014**

**YEAR 12 HSC ASSESSMENT TASK 1**

# Mathematics Extension 1

## General Instructions

- Reading time – 5 minutes
- Working time – 55 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work
- Attempt all questions

## Total marks – 36

This paper consists of 3 questions

## Questions 1-3 – Extended Response 36 marks

Start a new page for each question.

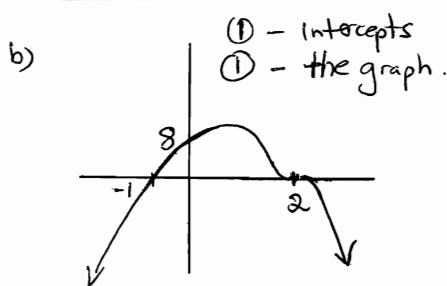
| Question 1 (17 marks) - Start a new page |                                                                                                                                                                                                                                                                                                                                                                   | Marks                                                           |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| a)                                       | <p>A polynomial <math>P(x) = ax^n(x - 1)^2 + 5x + 7</math> is monic of degree 6.</p> <p>(i) Find the value of <math>a</math>.</p> <p>(ii) Find the value of <math>n</math>.</p> <p>(iii) What is the remainder when <math>P(x)</math> is divided by <math>(x - 1)</math>?</p>                                                                                     | <p><b>1</b></p> <p><b>1</b></p> <p><b>1</b></p>                 |
| b)                                       | Sketch the graph of $y = (2 - x)^3(x + 1)$ showing the $x$ and $y$ intercepts.                                                                                                                                                                                                                                                                                    | <b>2</b>                                                        |
| c)                                       | Solve $x^3 + 3x^2 - 9x - 27 = 0$                                                                                                                                                                                                                                                                                                                                  | <b>3</b>                                                        |
| d)                                       | <p>If <math>\alpha, \beta</math> and <math>\gamma</math> are the roots of <math>2x^3 - 3x + 4x + 2 = 0</math>, find:-</p> <p>(i) <math>\alpha + \beta + \gamma</math></p> <p>(ii) <math>\alpha\beta\gamma</math></p> <p>(iii) <math>\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}</math></p> <p>(iv) <math>(\alpha - 1)(\beta - 1)(\gamma - 1)</math></p> | <p><b>1</b></p> <p><b>1</b></p> <p><b>2</b></p> <p><b>2</b></p> |
| e)                                       | <p>The polynomial <math>P(x) = x^3 + ax^2 + bx + 4</math> has <math>x + 2</math> as a factor. When <math>P(x)</math> is divided by <math>x - 1</math> the remainder is 9.</p> <p>Find the value of <math>a</math> and <math>b</math>.</p>                                                                                                                         | <b>3</b>                                                        |

| Question 2 (9 marks) - Start a new page |                                                                                                                                                    |   |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|---|
| a)                                      | Use mathematical induction to prove $\frac{1}{1.2} + \frac{1}{2.3} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$<br>where $n$ is a positive integer. | 3 |
| b)                                      | Two roots of the polynomial $2x^3 + x^2 - kx + 6 = 0$ are equal in absolute value but opposite in sign. Find the value of $k$ .                    | 3 |
| c)                                      | What is the remainder when $x^{99} - 99$ is divided by $x^2 - 1$ ?                                                                                 | 3 |

| Question 3 (10 marks) - Start a new page |                                                                                                                | Marks |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------|
| a)                                       | Use mathematical induction to prove $n^3 + 4n$ is divisible by 8 if $n$ is an even integer, where $n \geq 2$ . | 4     |
| b)                                       | Two of the roots of the equation $x^3 + ax^2 + b = 0$ are reciprocals of each other.                           |       |
|                                          | (i) Show that the third root is equal to $-b$                                                                  | 1     |
|                                          | (ii) Show that $a = b - \frac{1}{b}$                                                                           | 2     |
|                                          | (iii) Show that the 2 roots which are reciprocals will be real if $-\frac{1}{2} \leq b \leq \frac{1}{2}$       | 3     |

|  |                           |  |
|--|---------------------------|--|
|  | <b>End of Examination</b> |  |
|--|---------------------------|--|

- a) (i)  $a = 1$  ①  
 (ii)  $n = 4$  ①  
 (iii)  $P(1) = 12$  ①



c)  $P(x) = x^3 + 3x^2 - 9x - 27$   
 $P(3) = 27 + 27 - 27 - 27$   
 $\therefore (x-3)$  is a factor. ①

$$\begin{array}{r} x^2 + 6x + 9 \\ x-3 \overline{) x^3 + 3x^2 - 9x - 27} \\ \underline{x^3 - 3x^2} \phantom{- 9x - 27} \\ 6x^2 - 9x \phantom{- 27} \\ \underline{6x^2 - 18x} \phantom{- 27} \\ 9x - 27 \\ \underline{9x - 27} \\ 0 \end{array}$$

$\therefore P(x) = (x-3)(x+3)^2$  ①

$\therefore$  Solution  $x = 3, -3$  ①

d)  $2x^3 - 3x^2 + 4x + 2 = 0$

(i)  $\alpha + \beta + \gamma = \frac{3}{2}$  ①

(ii)  $\alpha\beta\gamma = -1$  ①

(iii)  $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$

$= \frac{2}{-1}$  ①

$= -2$  ①

(iv)  $(\alpha-1)(\beta-1)(\gamma-1)$

$= (\alpha-1)(\beta\gamma - \beta - \gamma + 1)$

$= \alpha\beta\gamma - \alpha\beta - \alpha\gamma + \alpha - \beta\gamma + \beta + \gamma - 1$

$= \alpha\beta\gamma - (\alpha\beta + \alpha\gamma + \beta\gamma) + \alpha + \beta + \gamma - 1$

$= -1 - 2 + \frac{3}{2} - 1$  ①

$= -\frac{5}{2}$  ①

e)  $P(x) = x^3 + ax^2 + bx + 4$

$P(-2) = 0 \therefore -8 + 4a - 2b + 4 = 0$

$4a - 2b = 4$

$2a - b = 2$  -- ①

$P(1) = 9 \quad 1 + a + b + 4 = 9$

$a + b = 4$  -- ②

$2a - b = 2$  -- ①

$a + b = 4$  -- ②

①+②  $3a = 6 \rightarrow a = 2$   
 $b = 2$

① mark for both eq's  
 ② marks for solving for "a" and "b". (1 mark off for each error)

### Question 2 [9]

a) Prove

$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$

Step 1 Prove true for  $n=1$

LHS =  $\frac{1}{2}$  RHS =  $\frac{1}{1+1} = \frac{1}{2}$

LHS = RHS  $\therefore$  true for  $n=1$ .

(accept  $\frac{1}{1 \cdot 2} = \frac{1}{1+1}$ )

Step 2 Assume true for  $n=k$

$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} = \frac{k}{k+1}$  ①

Step 3. Prove true for  $n=k+1$   
 i.e.

$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} = \frac{k+1}{k+2}$

from Assumption ①

$\frac{k}{k+1} + \frac{1}{(k+1)(k+2)} = \frac{k+1}{k+2}$

LHS =  $\frac{k(k+2) + 1}{(k+1)(k+2)}$

$= \frac{k^2 + 2k + 1}{(k+1)(k+2)}$

$= \frac{(k+1)^2}{(k+1)(k+2)}$

$= \frac{k+1}{k+2}$

= RHS

$\therefore$  Proven true for  $n=k+1$   
 if true for  $n=k \therefore$  true  
 for  $n=1, n=2$  & for all  
 $n$  by mathematical  
 induction.

① mark for steps 1, 2 and 4 correct

② marks for successfully proving step 3.

2b)  $2x^3 + x^2 - kx + 6 = 0$

$\alpha, -\alpha, \beta$

$\therefore \alpha + -\alpha + \beta = -\frac{1}{2}$

$\beta = -\frac{1}{2}$  -- ①

$\alpha\beta - \alpha\beta - \alpha^2 = -\frac{k}{2}$

$\alpha^2 = \frac{k}{2}$  -- ②

$-\alpha^2\beta = -\frac{6}{2}$

$\alpha^2\beta = 3$  -- ③

sub ① into ③

$\alpha^2(-\frac{1}{2}) = 3$

$\alpha^2 = -6$

sub into ②

$-6 = \frac{k}{2}$

$k = -12$

Alternatively sub  $\beta = -\frac{1}{2}$  back into equation & solve.

Marking scheme

① for the 3 equations

② marks for using eq's to solve for  $k$ .

c)  $P(x) = x^{99} - 99$

$P(x) = (x^2 - 1)Q(x) + ax + b$

$\therefore x^{99} - 99 = (x^2 - 1)(Q(x)) + ax + b$

Sub in  $x=1$

(1)<sup>99</sup> - 99 =  $a(1) + b$

$a + b = -98$  -- ①

sub in  $x=-1$

(-1)<sup>99</sup> - 99 =  $-a + b$

$-a + b = -100$  -- ②

①+②  $2b = -198$

$b = -99$

$a = 1$

$\therefore$  remainder is  $x - 99$

① for  $(x^2 - 1)Q(x) + ax + b$

① for equations ① + ②

① for solving successfully, but must write remainder  $x - 99$ .

3a) Prove  $n^3 + 4n$  is  $\div$  by 8 if  $n$  is even

Step 1. Prove true for  $n=2$

$$2^3 + 4(2) = 16$$

$$\frac{16}{8} = 2 \therefore \text{true for } n=2.$$

Step 2. Assume true for  $n=k$

$$\text{i.e. } \frac{k^3 + 4k}{8} = M \text{ (where } M \text{ is an integer)}$$

$$\text{i.e. } k^3 = 8M - 4k. \quad \text{---(1)}$$

Step 3. Prove true for  $n=k+2$ .

$$\text{i.e. } (k+2)^3 + 4(k+2) \text{ is } \div \text{ by } 8, k \text{ even.}$$

$$k^3 + 6k^2 + 16k + 16 \text{ is } \div \text{ by } 8$$

sub in (1)

$$\begin{aligned} & 8M - 4k + 6k^2 + 16k + 16 \\ &= 8M + 6k^2 + 12k + 16 \quad \text{---(2)} \end{aligned}$$

but  $k$  is even  $\therefore$   
let  $k=2a$  where  $a$  is a positive integer

$$\begin{aligned} &= 8M + 6(2a)^2 + 12(2a) + 16 \\ &= 8M + 24a^2 + 24a + 16 \end{aligned}$$

$$= 8(M + 3a^2 + 3a + 2)$$

$$\text{and since } 8 \text{ is a factor and } M \text{ and } a \text{ integers}$$

$(k+2)^3 + 4(k+2)$  is  $\div$  by 8  
step 4 proven true for  $k+2$   
if true for  $k$   $\therefore$   
true for 2, 4, 6, ...  $\therefore$   
for all even  $n$  by  
Mathematical Induction.

① mark for 1, 2 and 4

Step 3 - 3 marks

① for substituting  $k+2$ .

① for expanding  $(k+2)^3$  correctly & getting to expression ②

① for substituting  $k=2a$ .

NB If don't state  $M$  is an integer lose 1 mark

NB If make same error in step 4 as in earlier M.I. question 2a) don't deduct marks twice.

$$b) x^3 + ax^2 + b = 0$$

roots  $\alpha, \frac{1}{\alpha}, \beta$

$$(i) \alpha \times \frac{1}{\alpha} \times \beta = -b$$

$$\therefore \beta = -b \quad \text{① mark}$$

$$(ii) \alpha + \frac{1}{\alpha} - b = -a$$

$$\alpha + \frac{1}{\alpha} = b - a \quad \text{---(1)}$$

$$\alpha(\frac{1}{\alpha}) + \alpha(\beta) + \frac{\beta}{\alpha} = 0$$

$$\text{i.e. } 1 - \alpha\beta - \frac{b}{\alpha} = 0$$

$$1 - b(\alpha + \frac{1}{\alpha}) = 0 \quad \text{---(2)} \quad \text{① mark}$$

sub ① into ②

$$1 - b(b - a) = 0$$

$$-\frac{1}{b} = (b - a)$$

$$a = b - \frac{1}{b} \quad \text{---(3)} \quad \text{① mark}$$

$$\text{ciii) from ① } \alpha + \frac{1}{\alpha} = b - a$$

$$\text{sub in ③ } \alpha + \frac{1}{\alpha} = b - (b - \frac{1}{b})$$

$$\alpha + \frac{1}{\alpha} = b - (b - \frac{1}{b})$$

$$\alpha + \frac{1}{\alpha} = \frac{1}{b}$$

$$\alpha^2 b - \alpha + b = 0 \quad \text{---(4)}$$

$$\therefore \alpha = \frac{1 \pm \sqrt{1 - 4b^2}}{2b}$$

roots are real if

$$1 - 4b^2 \geq 0$$

$$(1 + 2b)(1 - 2b) \geq 0$$



$\therefore$  roots are real

$$\text{if } -\frac{1}{2} \leq b \leq \frac{1}{2}$$

1 mark for sub. in ③ into

① & getting eq'n ④

1 mark for using quad. form. correctly.

1 mark for correct conclusion using  $\Delta$ .